

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with Llama and IBM Bob”

Project Title – ECO-LIFESTYLE AGENT

Domain of Project – SUSTAINABILITY AND ENVIRONMENTALMANAGEMENT

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PROBLEM STATEMENT

Problem Statement No.6 – Eco Lifestyle Agent

The Challenge –

- ✓ An Eco Lifestyle Agent, powered by RAG (Retrieval-Augmented Generation), empowers users to adopt a greener lifestyle through personalized, practical suggestions.
- ✓ It retrieves sustainable living tips, eco-friendly product recommendations, local recycling guidelines, and government schemes from trusted environmental sources.
- ✓ Users can ask natural language questions such as “How can I reduce plastic use at home?” or “What are eco-friendly travel options in my city?” and receive instant, actionable guidance.
- ✓ The agent promotes small daily actions with big environmental impact, making sustainability easy and accessible. This AI-driven assistant fosters eco-conscious decisions, raises awareness, and helps build a more sustainable future.

Technology – Use of IBM Cloud Lite services / IBM Granite is mandatory.

PROPOSED SOLUTION

Proposed Solution: AI-Powered Eco Lifestyle Agent

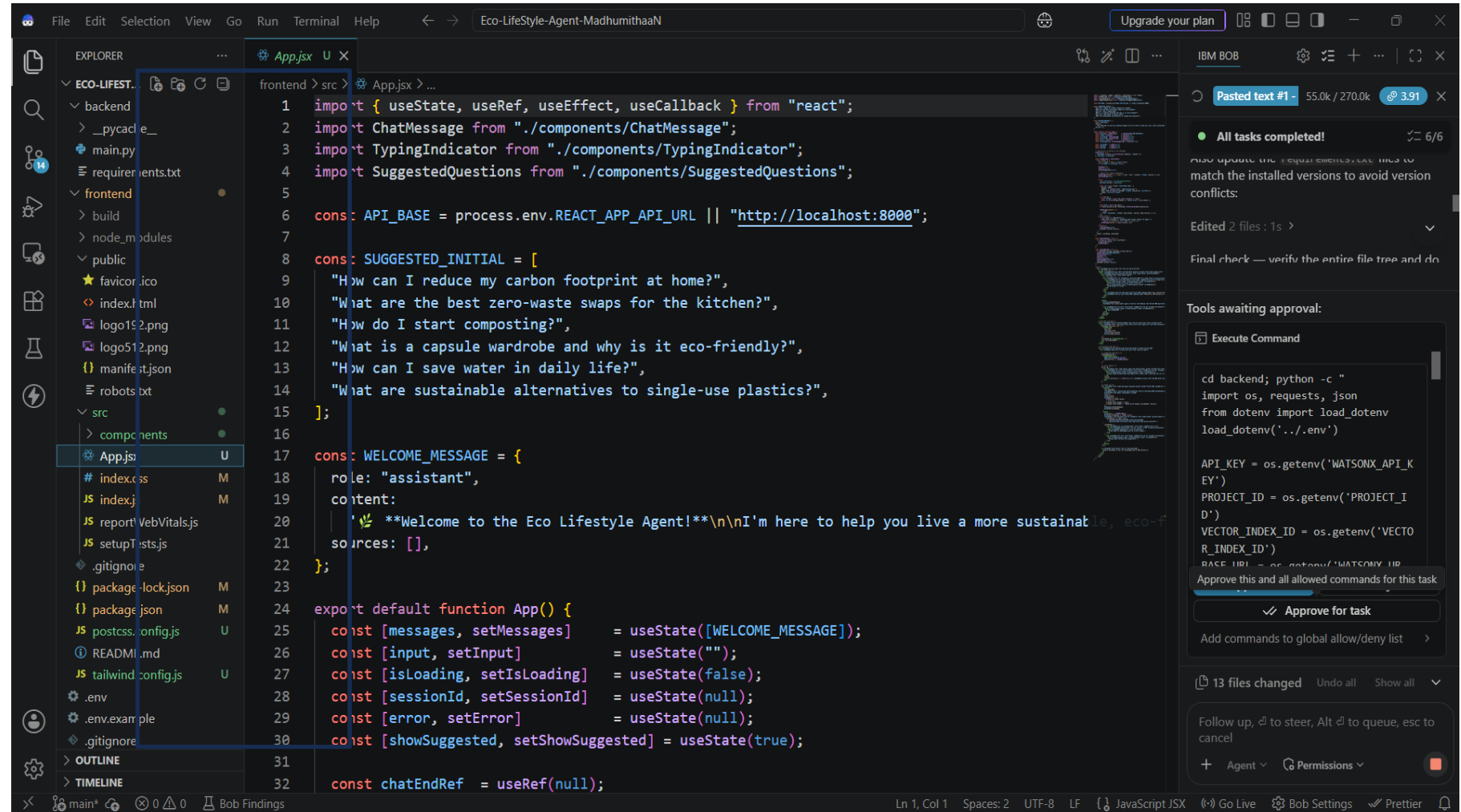
- ✓ The proposed solution is an AI-powered Eco Lifestyle Agent built using IBM watsonx.ai and the Llama-3-3-70b-instruct model to provide personalized, real-time sustainability guidance.
- ✓ The system addresses challenges in eco-conscious decision-making, waste reduction, and sustainable living through an intelligent, retrieval-grounded architecture.
- ✓ At its core, the solution uses a Retrieval-Augmented Generation (RAG) pipeline to fetch accurate sustainability information from a curated knowledge base covering eco-friendly practices, recycling guidelines, and government green schemes.
- ✓ *The system is organized around four core functional capabilities:*
 - i. **Eco Knowledge Retrieval:** Semantically searches the knowledge base to surface the most relevant sustainability content for any user query, using vector similarity search over pre-indexed documents.
 - ii. **Personalized Recommendation Engine:** Uses Llama-3-3-70b-instruct to generate tailored, actionable suggestions (zero-waste swaps, eco-friendly product alternatives, local recycling guidance) based on the user's specific question and context.
 - iii. **Sustainable Living Advisory:** Offers practical, category-specific guidance — from reducing plastic use at home to eco-friendly travel options — grounded strictly in retrieved source material rather than the model's unverified general knowledge.

TECHNOLOGIES USED

- ✓ **IBM Bob Platform** – For designing and orchestrating the agent workflow visually, coordinating the retrieval, embedding, and generation steps of the pipeline.
- ✓ **IBM watsonx.ai** – Provides the platform for model deployment, secure IAM-based authentication, and scalable inference, powering both the embedding and chat generation calls.
- ✓ **Llama-3-3-70b-instruct(via watsonx.ai)** – Powers natural language understanding, reasoning, and personalized response generation for the Eco Lifestyle Agent.
- ✓ **RAG (Retrieval-Augmented Generation)** – Retrieves relevant sustainability data (eco-friendly practices, recycling guidelines, government schemes) before generation, making responses source-grounded and trustworthy rather than relying on the model's raw knowledge.
- ✓ **Vector Index (watsonx.ai Data Asset)** – Stores pre-embedded knowledge base chunks and enables local similarity search (L2 distance) for fast, low-latency retrieval at query time.
- ✓ **IBM watsonx Orchestrate on IBM Cloud** – Provides the platform to build, deploy, and manage the Eco Lifestyle Agent efficiently, enabling reliable and scalable agent orchestration and interaction.

REQUIRED FILES CREATED BY IBM BOB

- ✓ index.html
- ✓ App.jsx
- ✓ index.css
- ✓ main.py
- ✓ requirements.txt



IBM BOB SCREENSHOTS

The screenshot displays the IBM BOB IDE interface. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. The Explorer panel on the left shows a project structure for 'Eco-LifeStyle-Agent-MadhumithaaN' with folders like backend, frontend, and src, and files like main.py, requirements.txt, and index.js. The main editor area shows a Python file named 'probe.py' with the following code:

```
1+ import os, requests, json
2+ from dotenv import load_dotenv
3+ load_dotenv('../.env')
4+
5+ API_KEY = os.getenv('WATSONX_API_KEY')
6+ PROJECT_ID = os.getenv('PROJECT_ID')
7+ VECTOR_INDEX_ID = os.getenv('VECTOR_INDEX_ID')
8+ BASE_URL = os.getenv('WATSONX_URL')
9+
10+ iam_resp = requests.post(
11+     'https://iam.cloud.ibm.com/identity/token',
12+     headers={'Content-Type': 'application/x-www
13+     data={'grant_type': 'urn:ibm:params:oauth:g
14+     timeout=30
15+ )
16+ token = iam_resp.json()['access_token']
17+ headers = {'Accept': 'application/json', 'Conte
18+
19+ # 1. List all available models
20+ r = requests.get(f'{BASE_URL}/ml/v1/foundation_
21+ d = r.json()
22+ resources = d.get('resources', [])
23+ total = d.get('total_count')
24+ print(f'Total models: {total}')
25+ for m in resources:
26+     funcs = m.get('functions', [])
27+     func_ids = [f.get('id', '') for f in funcs]
28+     print(f" {m['model_id']}<55} {func_ids}")
29+
30+ # 2. Try the IBM Watson Discovery v2 endpoint -
31+ print('\n--- Discovery probe ---')
32+ # The vector index ID format 01a07a6b-7c94-754d
```

The right sidebar contains the 'IBM BOB' panel, which shows a task list. The first task is 'Pasted text #1' with a status of 'All tasks completed!'. Below this, there is a section for 'Running 3 commands' with a list of commands and their results. The commands are:

- Ran cd backend; python -c " i...
- Ran cd backend; python -c " i...
- Command failed (Exit code: 1) ...

The bottom status bar shows the current file is 'main.py', the cursor is at line 1, column 1, and the file encoding is UTF-8. The status bar also includes a 'Go Live' button and a 'Bob Settings' link.

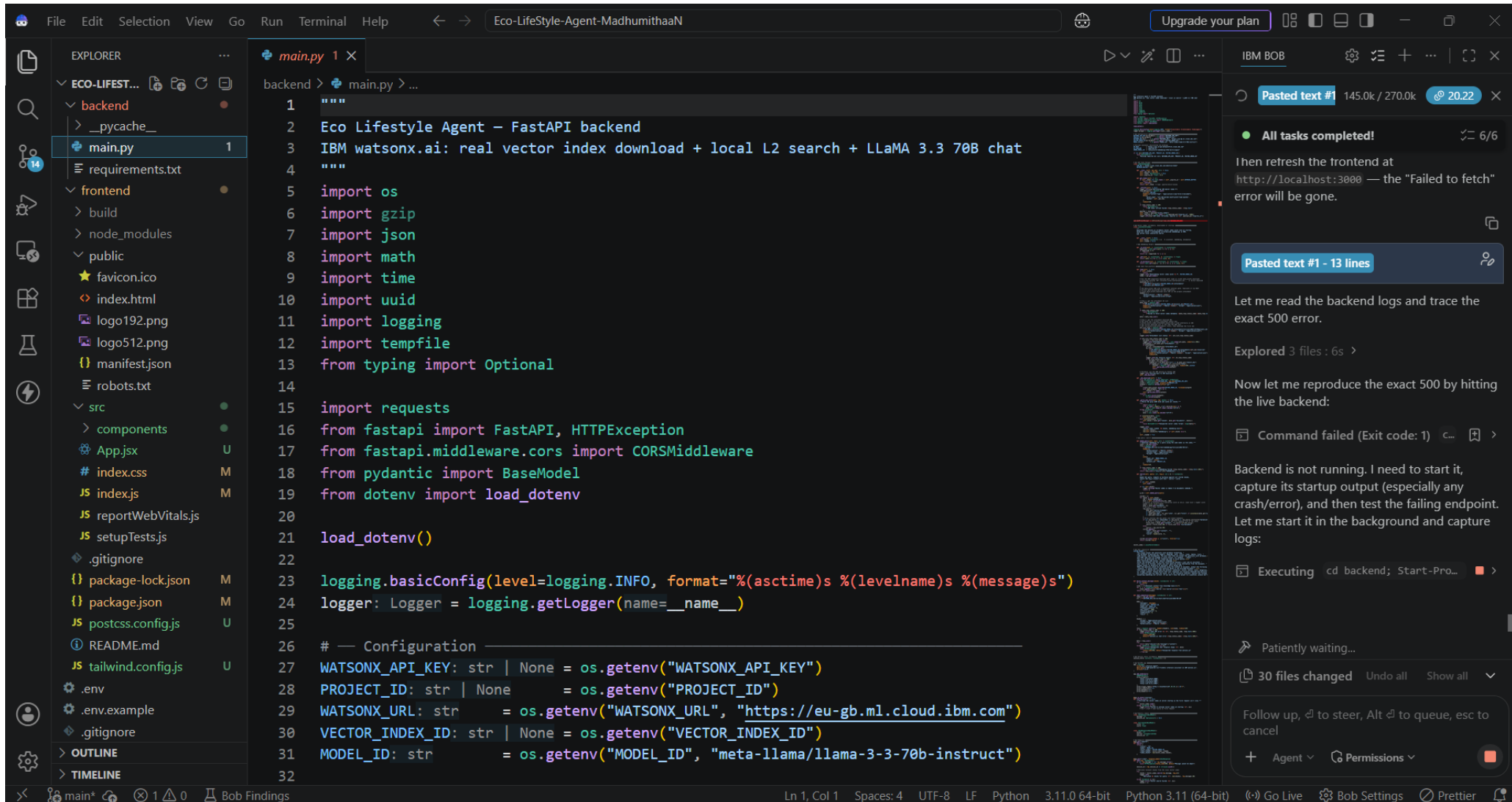
✓ index.css

The image shows a VS Code editor window with the following components:

- Explorer (Left):** Displays the project structure. The `src` folder is expanded, showing `components` and `App.jsx`. The `index.css` file is selected.
- Editor (Center):** Displays the content of `index.css`. The code is as follows:

```
1 @import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=...');
2 @tailwind base;
3 @tailwind components;
4 @tailwind utilities;
5
6 @layer base {
7   html {
8     @apply scroll-smooth;
9   }
10  body {
11    @apply bg-forest-50 text-gray-800 font-sans antialiased;
12  }
13  * {
14    @apply box-border;
15  }
16 }
17
18 @layer components {
19   .chat-bubble-user {
20     @apply bg-forest-600 text-white rounded-3xl rounded-br-sm px-4 py-3 shadow-bubble max-w-[80%] self-...
21   }
22   .chat-bubble-assistant {
23     @apply bg-white text-gray-800 rounded-3xl rounded-bl-sm px-4 py-3 shadow-bubble max-w-[80%] self-s...
24   }
25   .source-pill {
26     @apply inline-flex items-center gap-1 text-xs font-medium bg-bark-100 text-bark-700 border border-t...
27   }
28   .suggested-chip {
29     @apply inline-flex items-center text-xs font-medium bg-forest-50 text-forest-700 border border-fon...
30   }
31 }
32
```
- Right Sidebar:** Contains a task runner interface. It shows a task named "Pasted text #1" with a status of "All tasks completed!". Below this, it lists files being explored: `Read file frontend/package.json`, `Read file frontend/src/App.jsx (1, 15)`, and `Read file backend/main.py (360, 385)`. At the bottom, there is a section titled "Tools awaiting approval:" with a command box containing `netstat -ano | Select-String ":8000"` and `| Select-Object -First 5`. There are buttons for "Approve once", "Reject", and "Approve for task".

✓ main.py



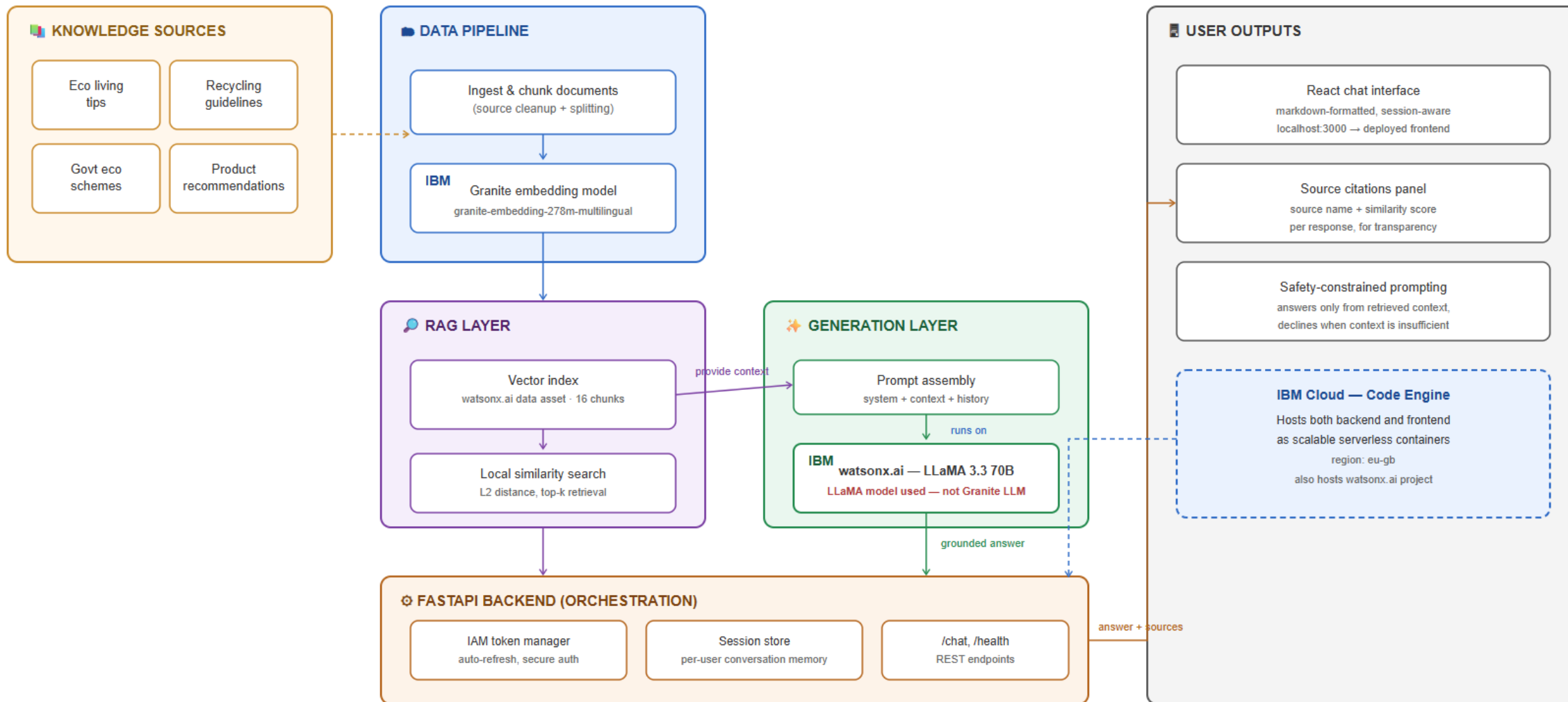
The screenshot shows a VS Code editor with a file named `main.py` open. The file is located in the `backend` directory. The code is a Python script for a FastAPI backend. It includes imports for `os`, `gzip`, `json`, `math`, `time`, `uuid`, `logging`, `tempfile`, `typing`, `requests`, `fastapi`, `fastapi.middleware.cors`, `pydantic`, and `dotenv`. It also includes a configuration section with environment variables for `WATSONX_API_KEY`, `PROJECT_ID`, `WATSONX_URL`, `VECTOR_INDEX_ID`, and `MODEL_ID`. The code is currently running in a terminal window, and the output shows a 500 error.

```
1 """
2 Eco Lifestyle Agent - FastAPI backend
3 IBM watsonx.ai: real vector index download + local L2 search + LLaMA 3.3 70B chat
4 """
5 import os
6 import gzip
7 import json
8 import math
9 import time
10 import uuid
11 import logging
12 import tempfile
13 from typing import Optional
14
15 import requests
16 from fastapi import FastAPI, HTTPException
17 from fastapi.middleware.cors import CORSMiddleware
18 from pydantic import BaseModel
19 from dotenv import load_dotenv
20
21 load_dotenv()
22
23 logging.basicConfig(level=logging.INFO, format="%(asctime)s %(levelname)s %(message)s")
24 logger: Logger = logging.getLogger(name=__name__)
25
26 # --- Configuration ---
27 WATSONX_API_KEY: str | None = os.getenv("WATSONX_API_KEY")
28 PROJECT_ID: str | None = os.getenv("PROJECT_ID")
29 WATSONX_URL: str = os.getenv("WATSONX_URL", "https://eu-gb.ml.cloud.ibm.com")
30 VECTOR_INDEX_ID: str | None = os.getenv("VECTOR_INDEX_ID")
31 MODEL_ID: str = os.getenv("MODEL_ID", "meta-llama/llama-3-3-70b-instruct")
32
```

The terminal output shows a 500 error:

```
INFO: 127.0.0.1:5000 - "POST /v1/chat/completions HTTP/1.1" 500 Internal Server Error
```

ARCHITECTURE BLUEPRINT FOR IBM BOB



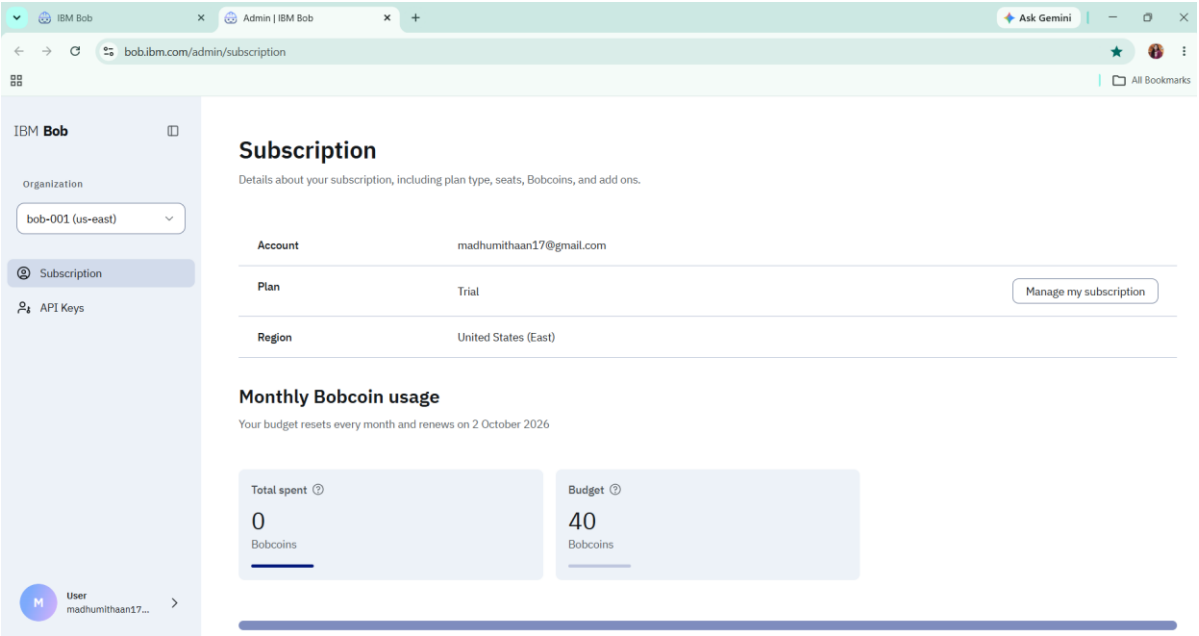
ROLE OF RETRIVAL AUGMENTED GENERATION(RAG) AND AI IN THE SOLUTION

- ✓ **Grounded, accurate responses** — RAG retrieves from a curated knowledge base of sustainability facts and guidelines before generation, so the watsonx.ai model's answers are anchored in verified sources rather than relying purely on the LLM's parametric memory.
- ✓ **Reduced hallucination on domain-specific facts** — for numeric or factual claims (emission figures, recycling guidelines), retrieval grounds the LLM's output in retrieved passages instead of the model guessing.
- ✓ **Dynamic, updatable knowledge without retraining** — new sustainability data, guidelines, or product info can be added to the retrieval index directly, so the system stays current without fine-tuning the underlying model.
- ✓ **Context-aware conversation** — the AI layer (via LangChain-style orchestration) combines retrieved context with conversation history, letting the chatbot give relevant answers, in your case eco-lifestyle advice, that stay on-topic across a multi-turn chat.

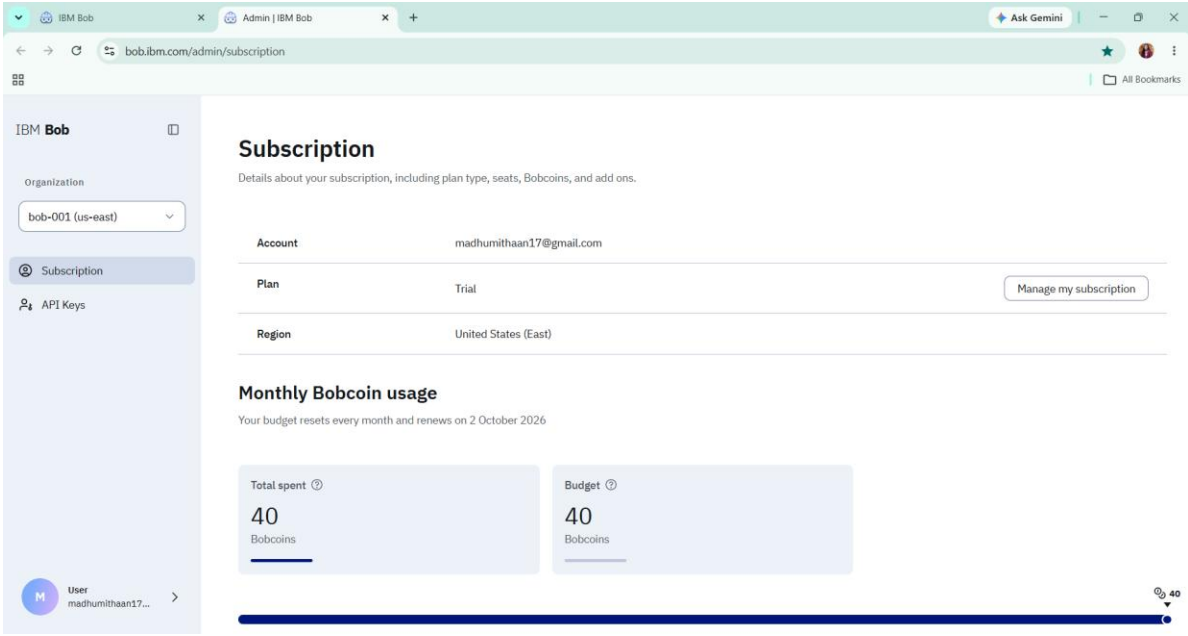


TOKEN USAGE

Before Prompt

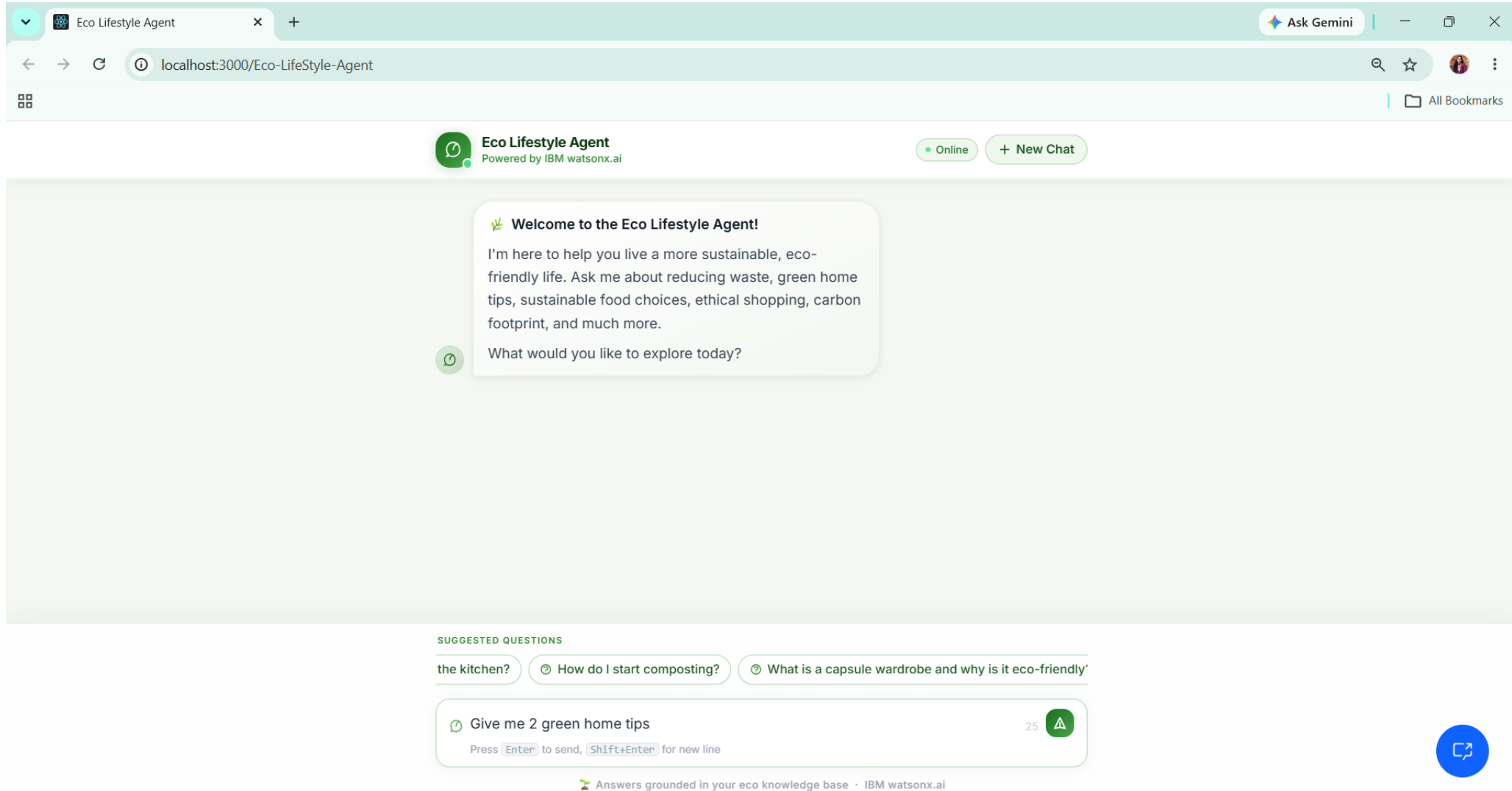


After the Prompt

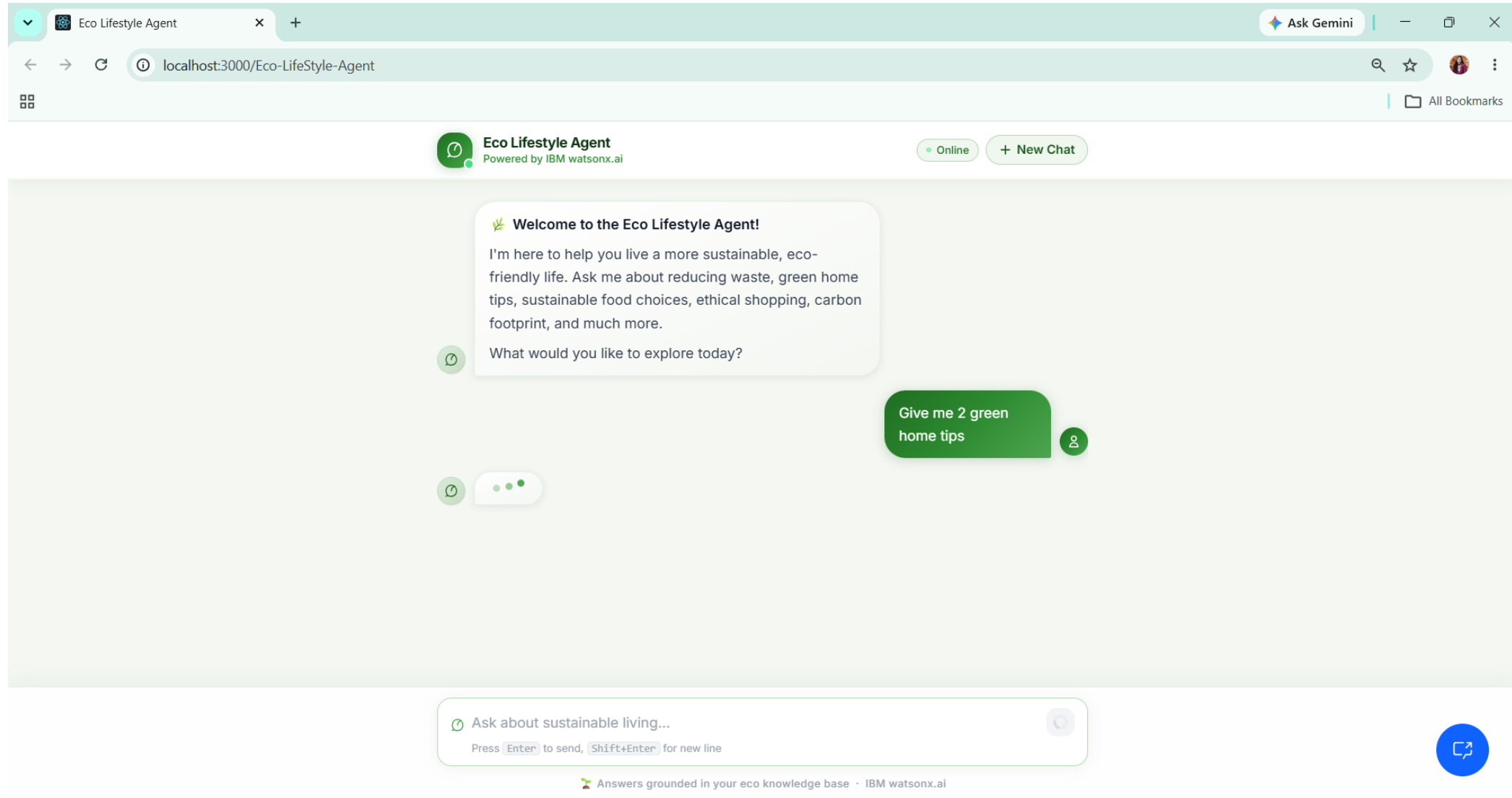


PROJECT OUTPUT SCREENSHOT

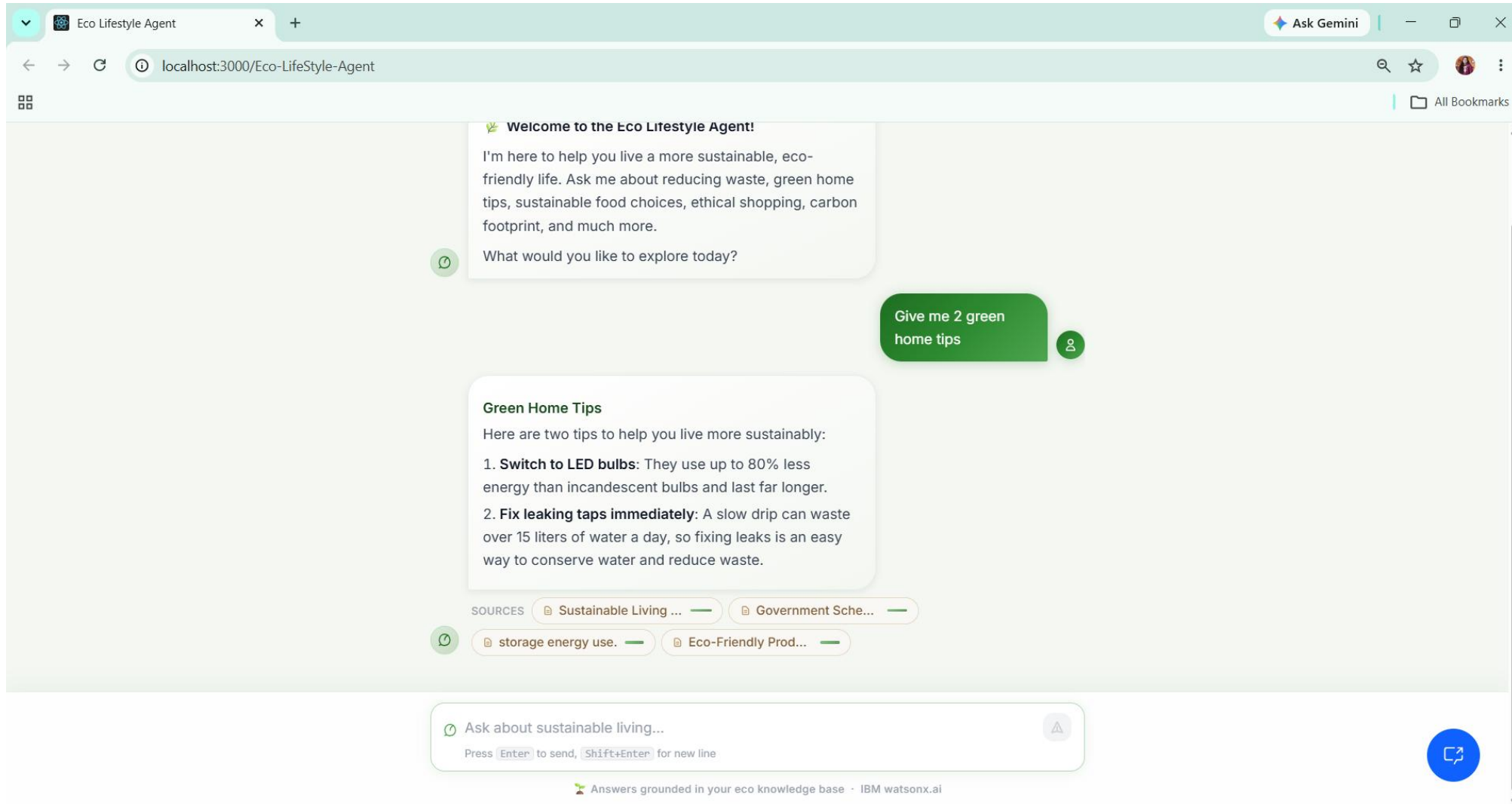
✓ **Agent Interface** – The user initiates a conversation with the Eco Lifestyle Agent.



- ✓ **Query Processing** – The agent processes and reasons over the user's request using its knowledge base.



- ✓ **Response Generation** – The agent delivers a relevant, personalized sustainability response to the user.



- ✓ ***When the user prompts the Eco Lifestyle Agent with a question such as ‘Give 2 ways for recycling everyday items’ the agent provides concise, practical, and relevant sustainability recommendations.***

The screenshot displays a web browser window with the address bar showing 'localhost:3000/Eco-LifeStyle-Agent'. The page title is 'Eco Lifestyle Agent' and it is powered by 'IBM watsonx.ai'. The interface includes a chat area with a welcome message from the agent and a user query. The agent's response provides two practical recycling tips: reusing glass jars and repurposing plastic items. The chat area also shows sources for the information and a footer with a copyright notice.

Eco Lifestyle Agent
Powered by IBM watsonx.ai

Online + New Chat

Welcome to the Eco Lifestyle Agent!
I'm here to help you live a more sustainable, eco-friendly life. Ask me about reducing waste, green home tips, sustainable food choices, ethical shopping, carbon footprint, and much more.
What would you like to explore today?

Give 2 ways for recycling everyday items.

Recycling Everyday Items
Two ways to recycle everyday items are:

- 1. Reuse glass jars:** Reuse glass jars from jams or pickles as storage containers instead of buying new plastic organizers.
- 2. Repurpose plastic items:** Repair plastic items (buckets, containers) instead of replacing them when possible, to reduce the amount of plastic waste generated.

SOURCES storage energy use. Say no to plastic ...

Ask about sustainable living...
Press Enter to send, Shift+Enter for new line

Answers grounded in your eco knowledge base · IBM watsonx.ai

- ✓ ***When the user prompts the Eco Lifestyle Agent with a question such as ‘Give 3 tips for shopping sustainably’ the agent provides concise and practical recommendations.***

The screenshot shows a web browser window with the title "Eco Lifestyle Agent". The address bar displays "localhost:3000/Eco-LifeStyle-Agent". The page content features a chat interface with a light green background. A welcome message from the agent is displayed in a white bubble, followed by a user prompt in a green bubble. The agent's response is shown in a white bubble, containing a section titled "Sustainable Shopping Tips" with a list of three items. At the bottom, there is a text input field with a placeholder "Ask about sustainable living..." and a blue circular button with a share icon. A footer at the very bottom states "Answers grounded in your eco knowledge base · IBM watsonx.ai".

Welcome to the Eco Lifestyle Agent!
I'm here to help you live a more sustainable, eco-friendly life. Ask me about reducing waste, green home tips, sustainable food choices, ethical shopping, carbon footprint, and much more.

What would you like to explore today?

Give 3 tips for shopping sustainably.

Sustainable Shopping Tips
Here are three tips for shopping sustainably:

- 1. Buy in bulk or from refill stores:** This helps to cut down on single-use packaging, reducing waste and the environmental impact of packaging production.
- 2. Use reusable shopping bags:** Opt for cloth or jute bags that can be reused for years, reducing the need for single-use plastic bags. These bags are often foldable and can fit in a pocket or purse.
- 3. Choose products with minimal packaging:** When possible, select products with little to no plastic packaging, such as shampoo and soap bars, which often come with minimal or biodegradable packaging and can be more concentrated, lasting longer per gram.

Ask about sustainable living...

Press **Enter** to send, **Shift+Enter** for new line

Answers grounded in your eco knowledge base · IBM watsonx.ai

NOVELTY AND UNIQUENESS

- ✓ The proposed Eco-LifeStyle Agent combines Retrieval-Augmented Generation (RAG) with personalization to deliver tailored, context-aware sustainability guidance to users.
- ✓ **Grounded, source-cited sustainability guidance:** Unlike generic chatbots that answer from a model's general training data, every response is retrieved from a curated eco-knowledge base first, with the source and similarity score visible — turning vague eco-advice into traceable, verifiable guidance.
- ✓ **Conversational memory for iterative eco-planning:** Session-aware history lets users refine their situation naturally ("what about for a 2-person apartment?") instead of restating full context every time, mirroring how a real sustainability consultant would converse.
- ✓ **Combines authoritative and hyperlocal information in one interface:** By blending general environmental best-practice content with location-relevant elements (like local recycling rules and government schemes), it bridges the gap between "what should I do" and "what applies to me specifically" something most sustainability apps don't unify.
- ✓ **Constrained, honest-refusal prompting:** The system is explicitly instructed to say when it *can't* answer from the available knowledge base, rather than hallucinating plausible-sounding eco-tips — a meaningful trust differentiator in a space prone to greenwashing and misinformation.

GITHUB LINK

Repository Link : <https://github.com/Madhumithaa-N/Eco-LifeStyle-Agent.git>

The screenshot shows a web browser displaying the GitHub repository page for 'Madhumithaa-N/Eco-LifeStyle-Agent'. The browser's address bar shows the URL 'github.com/Madhumithaa-N/Eco-LifeStyle-Agent'. The repository page is divided into two main sections: a file list on the left and a sidebar on the right.

File List:

File/Folder Name	Language	Time
.bob	eco-agent	3 hours ago
backend	eco-agent	3 hours ago
frontend	eco-agent	3 hours ago
.env.example	eco-agent	3 hours ago
.gitignore	eco-agent	3 hours ago
01_sustainable_living_tips.pdf	Files-RAG	2 hours ago
02_reducing_plastic_use_at_home.pdf	Files-RAG	2 hours ago
03_eco_friendly_travel_options.pdf	Files-RAG	2 hours ago
04_local_recycling_guidelines_india.pdf	Files-RAG	2 hours ago
05_government_schemes_india.pdf	Files-RAG	2 hours ago
06_eco_friendly_product_recommendations.pdf	Files-RAG	2 hours ago
Dockerfile	eco-agent	3 hours ago
Problem Statement-Madhu.pdf	Problem-Statement	8 minutes ago
README.md	eco-agent	3 hours ago
requirements.txt	eco-agent	3 hours ago
uvicorn_err.txt	eco-agent	3 hours ago
uvicorn_out.txt	eco-agent	3 hours ago

Sidebar:

- through an intelligent, retrieval-grounded architecture**
- Readme**
- Activity**
- 0 stars**
- 0 watching**
- 0 forks**
- Releases**
 - No releases published
 - [Create a new release](#)
- Packages**
 - No packages published
 - [Publish your first package](#)
- Contributors 1**
 - Madhumithaa-N** Madhumithaa N
- Languages**
 - Python 97.1%
 - Dockerfile 2.9%

FUTURE SCOPE

- ✓ **Personalized recommendation engine** — extend the chatbot to track a user's habits over time (diet, commute, energy use) and generate tailored sustainability suggestions instead of generic answers.
- ✓ **Integration with IoT/smart home data** — pull live data from smart meters, fitness trackers, or vehicle apps to give real-time carbon footprint estimates rather than relying on self-reported inputs.
- ✓ **Multi-modal input support** — let users upload receipts, utility bills, or photos of packaging so the RAG pipeline can extract and reason over that data automatically.
- ✓ **Gamification and community features** — add leaderboards, streaks, or challenges (e.g. "reduce plastic use this week") to drive sustained behavior change rather than one-off queries.
- ✓ **Localization and regional knowledge bases** — expand the retrieval corpus with region-specific sustainability data (local recycling rules, region-specific carbon grids, local eco-friendly vendors) so advice is actionable wherever the user is

CERTIFICATES EARNED

In recognition of the commitment to achieve
professional excellence



Madhumithaa Narayan

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Aug 31, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/a5173582-38df-454f-86e0-3a0b1918a66d>



CERTIFICATES EARNED

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to
Madhumithaa Narayan

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 01 Sep 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.